PES UNIVERSITY - Dept of Computer Science & Engineering

Lab 4: Vibe Coding — Galaga Repair & Enhancements

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Assigned Topic: 31_galaga | **Date:** October 2026

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1. Project Overview & Objectives

This project enhances the Galaga-lite arcade game built in Python using Pygame. The game features curved enemy entry paths (Bézier curves), formation swaying, diving attack patterns, player shooting, scoring, and wave-based difficulty. The objective of Lab 4 is to use an LLM (Gemini / Antigravity) as an interactive pair-programming partner to diagnose and repair an intentional trajectory bug and implement three new gameplay features across iterative Git commits.

2. Task Implementation & Solutions

Task 1 — Fix Entry/Dive Bézier Path Bug:
Analysis: Cubic Bézier curves follow the standard equation B(t) = (1-t)³P■ + 3(1-t)²t P■ + 3(1-t)t² P■ + t³P■. In the starter code, the third weighted term was 3 * u * t * p2 instead of 3 * u * t * t * p2, missing the squared exponent on *t*. This caused enemies to noticeably overshoot and bulge awkwardly mid-flight.
Fix: Corrected the exponent to 3 * u * t * t * p2[0] and 3 * u * t * t * p2[1]. Enemies now follow smooth, continuous arcs into formation.

Task 2 — Implement enemy_tint(kind):
Overrode enemy coloration per enemy type (boss, red, blue) to give distinct arcade aesthetics with high visual contrast (Neon Emerald for Boss, Arcade Crimson for Red, Electric Azure for Blue).

Task 3 — Implement on_wave_start(wave):
Added an animated stage announcement banner (STAGE X - GET READY) rendered centered on screen for 2.0 seconds at the start of each wave.

Task 4 — Implement shield_charges(wave):
Calculates player energy shield hit absorption: 1 + (wave - 1) // 3 charges per wave. Rendered a glowing cyan energy shield circle around the player ship and added a real-time Shield counter on the HUD.

3. Video & Repository Deliverables

Deliverable	File / Link	Status
10s Video Before Changes	before_edit.mp4	Completed & Verified
10s Video After Changes	FINAL.mp4	Completed & Verified
Full Game Source Code	31_galaga/game.py	Tested & Working
Documentation & Lab README	README.md	Completed
Complete PDF Report	Galaga_Repair_Enhancements.pdf	Completed

4. LLM Chat History & Prompting Trajectory

Prompt 1: Cloned assigned repo (SETAPESU26/31_galaga) and verified baseline game.
Prompt 2: Identified Bézier formula bug in bezier() and recorded before_edit.mp4.
Prompt 3: Implemented all 4 tasks with separate Git commits.
Prompt 4: Recorded FINAL.mp4 and generated submission repository matching course structure.